

IBM APPLIED DATA SCIENCE CAPSTONE

Will the First Stage Land?

Predicting SpaceX Falcon 9 landing success to price a launch

Kautzar

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GitHub repository: <https://github.com/kautzarichramsyah/Applied-Data-Science-Capstone>

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Executive Summary

90

Falcon 9 launches analysed
(2010–2020)

66.7%

first stages that landed
successfully

83.3%

test accuracy (15 of 18
launches)

84.4%

SVM accuracy across 50
data splits

Methods: collected launch data from the SpaceX REST API and Wikipedia, cleaned it with pandas, explored it with SQL, seaborn, Folium and a Plotly Dash app, then tuned four classifiers with GridSearchCV.

What drives landing success: experience. Success rose from 0% (2010–2013) to 84–90% (2019–2020), and newer boosters (FT, B4, B5) land far more often than v1.0/v1.1.

Where and what: KSC LC-39A has the best record (76.9%); ES-L1, GEO, HEO and SSO orbits landed every time, GTO only 51.9%.

Prediction: all four tuned classifiers reach 83.3% on the 18-launch test set; across 50 resampled splits SVM is the best and most stable model (84.4%). Errors are false positives.

Introduction

Space Y wants to bid against SpaceX — so it needs to know what a Falcon 9 launch really costs

BACKGROUND

SpaceX advertises Falcon 9 launches at about \$62 million; other providers charge upwards of \$165 million. Most of the saving comes from landing and reusing the first stage.

WHY IT MATTERS

If we can predict whether the first stage will land, we can estimate the true cost of a launch — the information a competing bidder needs.

Questions we set out to answer

- How do payload mass, orbit, launch site and flight number affect landing success?
- Has landing success improved over time, and with which booster versions?
- Can a machine learning model predict a successful landing — and how reliably?

01

Methodology

Data collection, wrangling, EDA, interactive analytics and classification

Methodology

Executive summary of the methodology

- 1 Data collection:** SpaceX REST API (requests + json_normalize) and web scraping of Wikipedia (BeautifulSoup)
- 2 Data wrangling:** Filter Falcon 9, fill missing payload with the mean, convert 8 landing outcomes into the Class label (1 = landed)
- 3 EDA:** Visualization (seaborn scatter, bar and line charts) and SQL queries on SQLite
- 4 Interactive visual analytics:** Folium maps (markers, clusters, distance lines) and a Plotly Dash dashboard
- 5 Predictive analysis:** Standardize, split 80/20, tune LR / SVM / Decision Tree / KNN with GridSearchCV (cv=10), evaluate on the test set and with repeated cross-validation

Data Collection

Two complementary sources for 2010–2020 Falcon 9 launches

SPACEX REST API

GET `api.spacexdata.com/v4/launches/past` (IBM static snapshot for consistent results). IDs for rocket, launchpad, payload and core are resolved with extra API calls to get booster version, site coordinates, payload mass, orbit, landing outcome, reuse and block.

WIKIPEDIA WEB SCRAPING

"List of Falcon 9 and Falcon Heavy launches" (9 June 2021 revision). HTML launch tables are parsed with BeautifulSoup into flight number, date, booster, site, payload, orbit, customer and outcomes.

Result: 90 Falcon 9 launches × 17 columns (API) and 121 scraped launch records, saved as CSV for wrangling.

Cleaning: Falcon 1 launches removed; 5 missing PayloadMass values replaced with the mean; 26 missing LandingPad values kept (no landing pad used).

Data Collection – SpaceX API

1

GET request

Launch data from the SpaceX REST API



2

Normalize

response.json() →
pd.json_normalize()



3

Enrich

getBoosterVersion,
getLaunchSite,
getPayloadData,
getCoreData



4

Filter

Keep Falcon 9 only
(drop Falcon 1) → 90
launches



5

Clean

Fill missing
PayloadMass with the
mean; export CSV

Key code

```
response = requests.get(static_json_url)           # status 200
data = pd.json_normalize(response.json())
getBoosterVersion(data); getLaunchSite(data); getPayloadData(data); getCoreData(data)
data_falcon9 = df[df['BoosterVersion'] != 'Falcon 1']
data_falcon9['PayloadMass'] = data_falcon9['PayloadMass'].fillna(data_falcon9['PayloadMass'].mean())
```

Data Collection – Scrapping

1

Request

GET the Wikipedia
Falcon 9 launch list
page



2

Parse

BeautifulSoup(respon
se.text,
"html.parser")



3

Find tables

soup.find_all("table")
→ launch tables



4

Extract

Column names from
<th>; rows from <td>



5

Save

launch_dict →
DataFrame → CSV

WHY A SECOND SOURCE

Wikipedia adds human-readable mission details
(customer, launch outcome, booster landing) that
cross-check the API records.

CHECK

soup.title returns "List of Falcon 9 and Falcon Heavy
launches - Wikipedia", confirming the right page was
parsed; 121 launch rows extracted.

Data Wrangling

Turning eight landing outcomes into one binary label

Outcome	Meaning	Count	Class
True ASDS	Landed on drone ship	41	1
None None	No landing attempted	19	0
True RTLS	Landed on ground pad	14	1
False ASDS	Drone ship landing failed	6	0
True Ocean	Controlled ocean landing	5	1
False Ocean	Ocean landing failed	2	0
None ASDS	Drone ship, no attempt	2	0
False RTLS	Ground pad landing failed	1	0

PROCESS

value_counts() of LaunchSite, Orbit and Outcome; bad_outcomes = outcomes 1, 3, 5, 6, 7; Class = 0 if outcome in bad_outcomes else 1.

RESULT

Mean of Class = 0.667: two out of three Falcon 9 first stages landed. CCAFS SLC 40 hosted 55 of the 90 launches; GTO is the most common orbit (27).

EDA with Data Visualization

Charts plotted and why

Chart	Variables	Why
Scatter (catplot)	Flight number vs. launch site, coloured by Class	Does experience at a site improve landings?
Scatter (catplot)	Payload mass vs. launch site	Do heavy payloads land less often? Site payload limits
Bar chart	Success rate by orbit type	Compare landing success across orbits
Scatter (catplot)	Flight number vs. orbit type	Does the experience effect differ by orbit?
Scatter (catplot)	Payload mass vs. orbit type	Payload-orbit interaction
Line chart	Yearly average success rate	Trend of landing success over time

Then: one-hot encoding of Orbit, LaunchSite, LandingPad and Serial → features_one_hot (90 × 80), cast to float64 for modelling.

EDA with SQL

Spacex.csv loaded into SQLite table SPACEXTABLE; queries performed

- Displayed the names of the unique launch sites (SELECT DISTINCT)
- Displayed 5 records where launch sites begin with 'CCA' (LIKE, LIMIT)
- Total payload mass carried by boosters launched by NASA (CRS) (SUM)
- Average payload mass carried by booster version F9 v1.1 (AVG)
- Date of the first successful landing on a ground pad (MIN)
- Boosters with drone-ship success and payload between 4,000 and 6,000 kg (AND, ranges)
- Total number of successful and failed mission outcomes (GROUP BY, COUNT)
- Booster versions that carried the maximum payload mass (subquery with MAX)
- 2015 drone-ship landing failures with month, booster and site (substr on Date)
- Ranking of landing outcomes between 2010-06-04 and 2017-03-20 (GROUP BY, ORDER BY DESC)

Build an Interactive Map with Folium

Map objects created and why

Map object	Purpose
folium.Circle + folium.Marker (DivIcon)	Mark and label each of the 4 launch sites at its latitude / longitude
MarkerCluster	Group the 56 launch records per site so overlapping launches stay readable
Green / red folium.Marker icons	Show each launch outcome (class 1 = landed, 0 = failed) at its site
MousePosition	Read coordinates of nearby coastline, towns, railways and highways
folium.PolyLine + distance label	Draw lines from CCAFS SLC-40 to its proximities and show the haversine distance

Questions answered: are sites near the Equator and the coast? Which site has the best outcome ratio? How far are sites from cities and transport links?

GitHub notebook: https://github.com/kautzarichramsyah/Applied-Data-Science-Capstone/blob/main/6_launch_site_location_folium.ipynb

Build a Dashboard with Plotly Dash

Plots and interactions added and why

Component	What it does	Why
Dropdown (site-dropdown)	All Sites or one of 4 launch sites	Compare sites or focus on one
Pie chart (success-pie-chart)	All sites: share of successes per site. One site: success vs. failure	See which site contributes most and which has the best rate
Range slider (payload-slider)	0–10,000 kg in 1,000 kg steps	Filter launches by payload
Scatter (success-payload-scatter-chart)	Payload vs. class, coloured by booster version	Find payload ranges and boosters with the most success

Two callbacks: the pie chart takes the dropdown as Input; the scatter takes both the dropdown and the slider as Inputs.

GitHub notebook: https://github.com/kautzarichramsyah/Applied-Data-Science-Capstone/blob/main/spacex_dash_app.py

Predictive Analysis (Classification)

How the classification models were built, tuned, evaluated and the best one selected



Model	Hyperparameters searched
Logistic regression	$C \in \{0.01, 0.1, 1\}$, penalty = l2, solver = lbfgs
Support vector machine	kernel $\in \{\text{linear, rbf, poly, sigmoid}\}$, C and gamma $\in \text{logspace}(-3, 3, 5)$
Decision tree	criterion, splitter, max_depth 2-18, min_samples_leaf, min_samples_split
K-nearest neighbours	n_neighbors 1-10, algorithm, p $\in \{1, 2\}$

Results

Summary of exploratory analysis, interactive analytics and predictive results

Exploratory data analysis

- Landing success rose from 0% (2010–2013) to 90% (2019)
- Overall success rate 66.7% across 90 launches
- ES-L1, GEO, HEO, SSO: 100%; GTO: 52%
- Newer boosters (FT, B4, B5) land far more often

Interactive analytics

- All sites are on the coast, close to the Equator
- KSC LC-39A: most successes (41.7%) and best rate (76.9%)
- CCAFS SLC-40 is 0.87 km from the coastline
- 2,000–5,000 kg payloads land most often

Predictive analysis

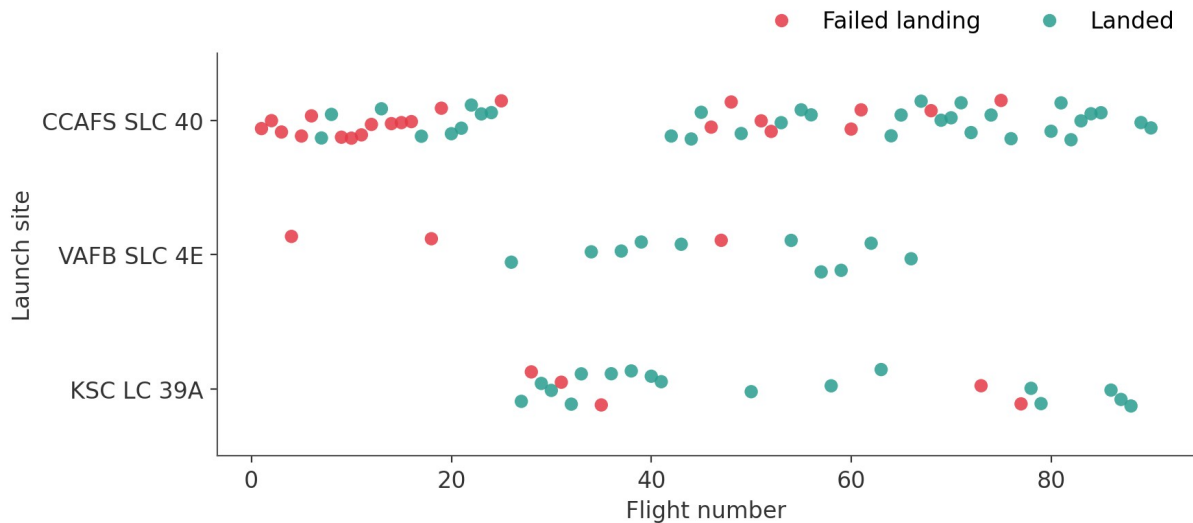
- All 4 models: 83.3% test accuracy (15 of 18)
- 100% recall, 80% precision; errors are false positives
- SVM is best over 50 splits: $84.4\% \pm 8.0\%$
- Decision tree overfits (75.1%)

02

Insights Drawn from EDA

EDA with visualization and EDA with SQL

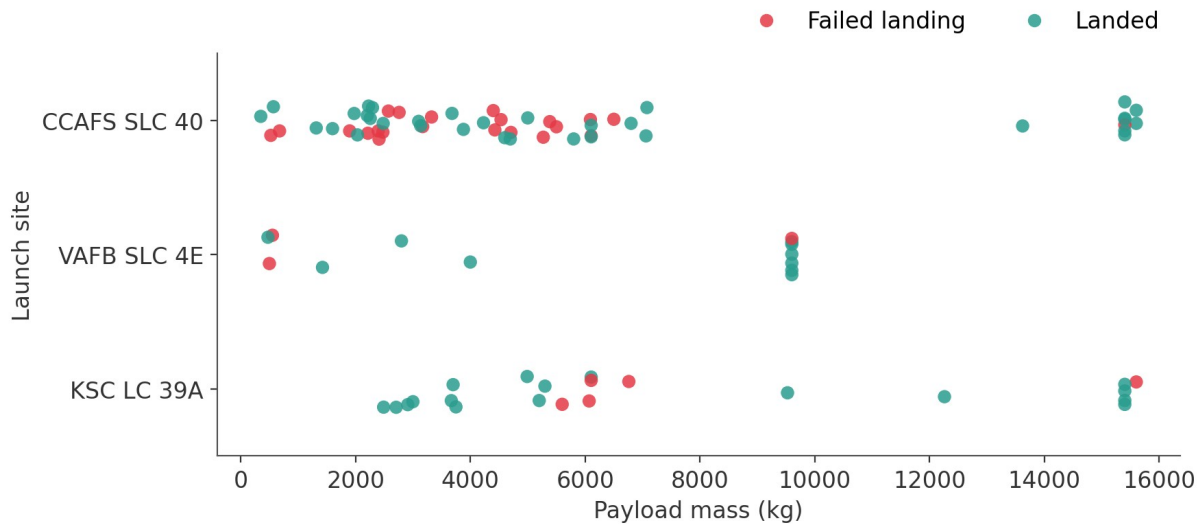
Flight Number vs. Launch Site



INSIGHT

Early flights (mostly from CCAFS SLC 40) failed to land. As flight numbers grow, green dominates at every site — experience matters more than location. KSC LC 39A only joined after flight 26 and started strong.

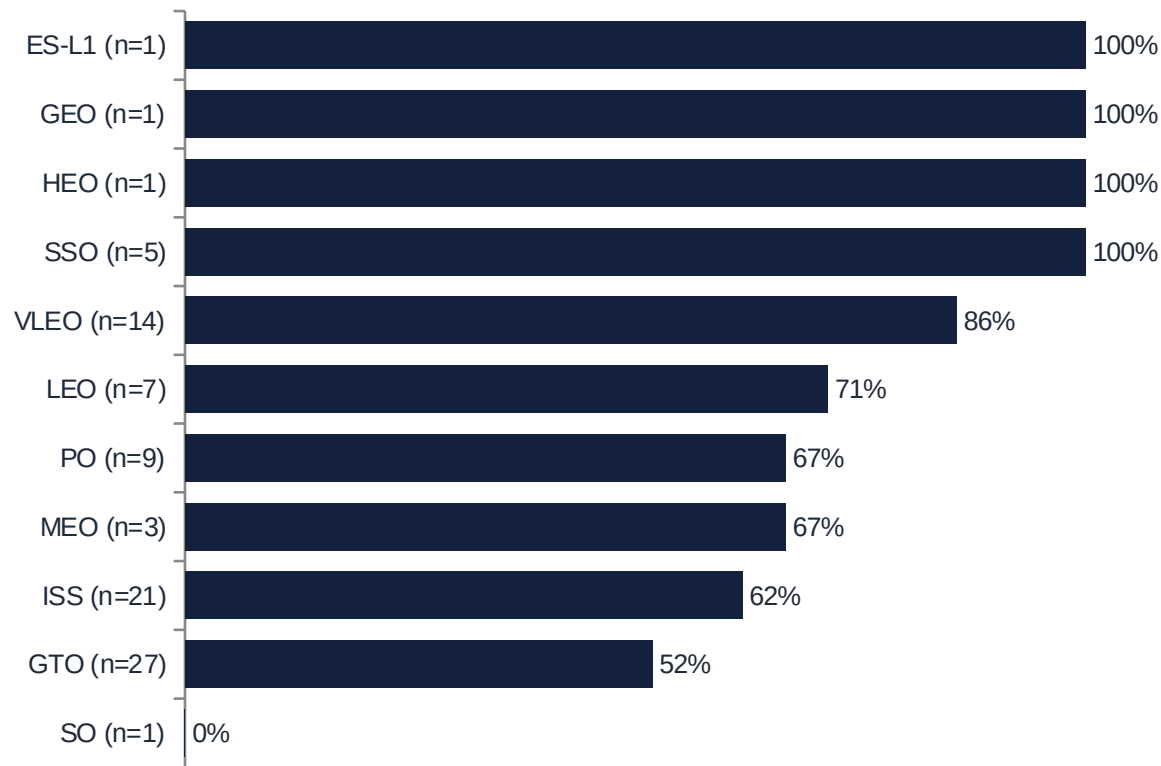
Payload vs. Launch Site



INSIGHT

VAFB SLC 4E launched no payloads above 10,000 kg. Heavy payloads (> 8,000 kg) mostly landed (87%), largely because they flew late on Block 5 boosters.

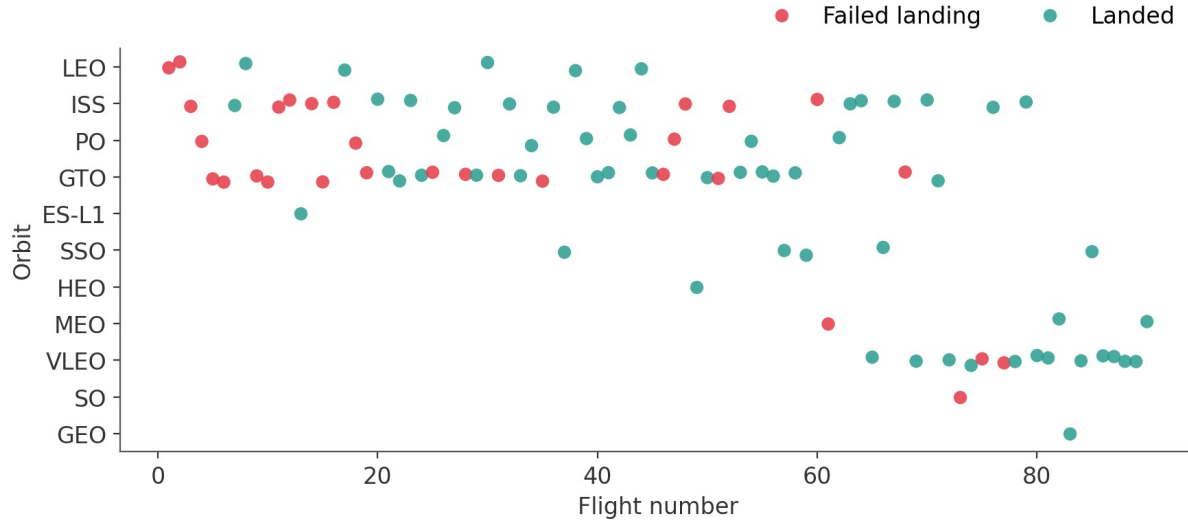
Success Rate vs. Orbit Type



INSIGHT

ES-L1, GEO, HEO and SSO landed every time, but with 1-5 launches each. Among the busy orbits, VLEO (86%) and LEO (71%) beat ISS (62%) and GTO (52%), the hardest orbit to recover from.

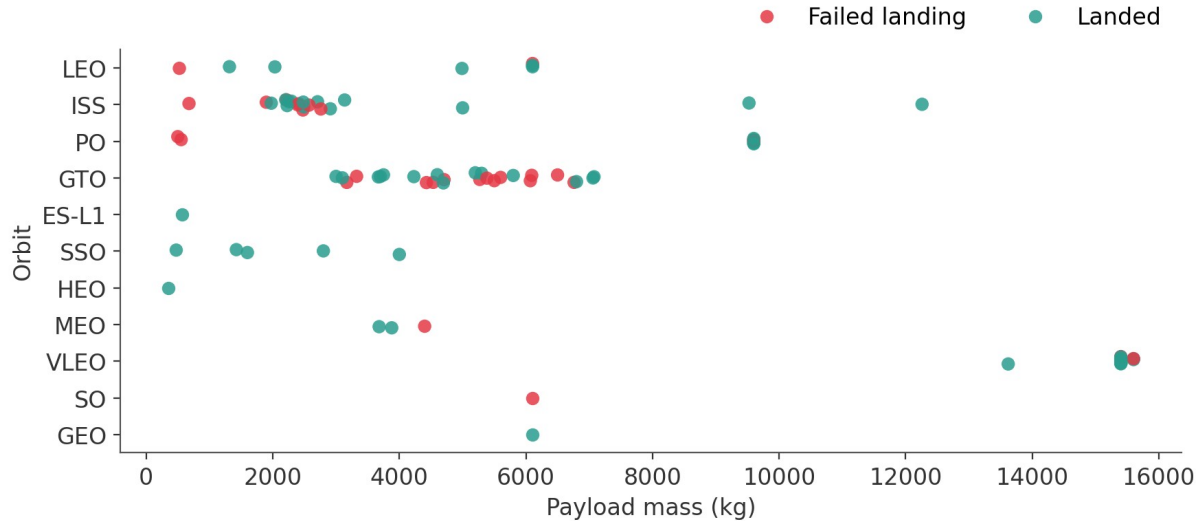
Flight Number vs. Orbit Type



INSIGHT

In LEO, success clearly tracks flight number. In GTO there is no such trend: failures continue into the later flights, so orbit energy, not just experience, limits recovery.

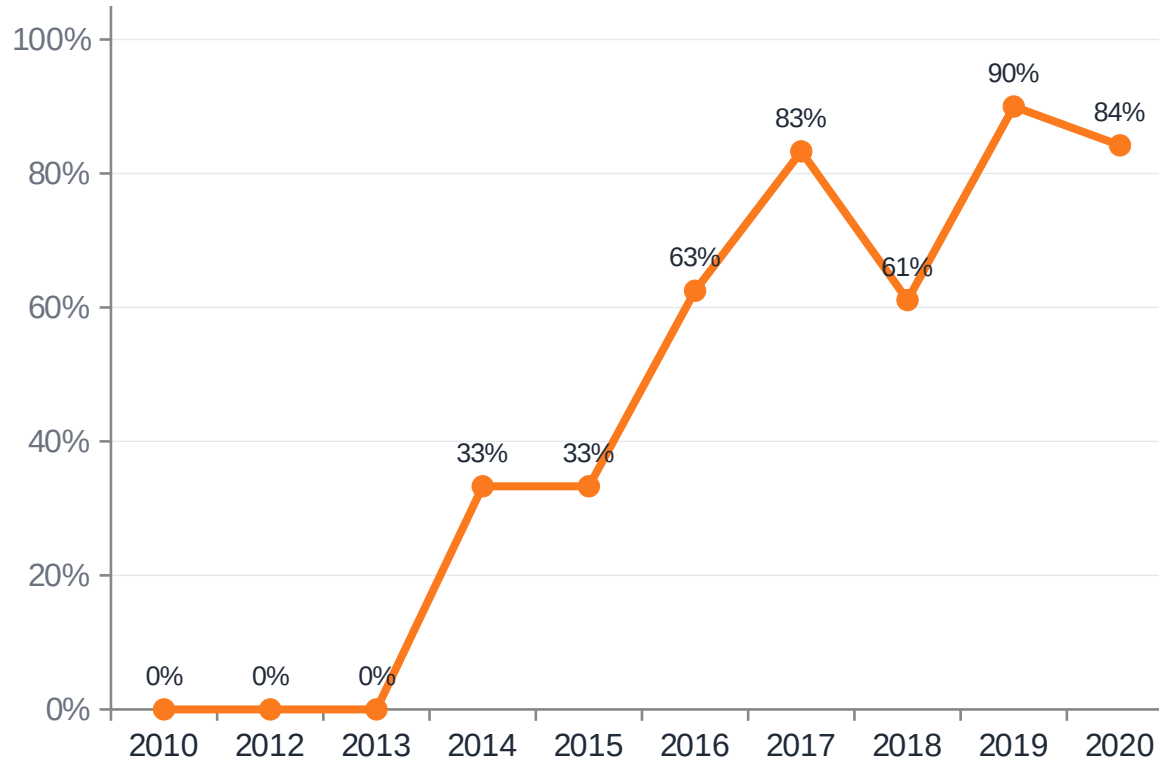
Payload vs. Orbit Type



INSIGHT

Heavy payloads land well in Polar, LEO, ISS and VLEO orbits. GTO payloads (3,000–7,000 kg) are a mix of successes and failures, with no clear payload threshold.

Launch Success Yearly Trend



INSIGHT

No landings succeeded from 2010 to 2013. The rate climbed steadily from 2013, dipped in 2018 as Block 5 was introduced, and reached 90% in 2019.

All Launch Site Names

Query `SELECT DISTINCT Launch_Site FROM SPACEXTABLE`

Launch_Site
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40

EXPLANATION

DISTINCT returns each launch site once. SpaceX used four sites: three in Florida (CCAFS LC-40, CCAFS SLC-40, KSC LC-39A) and one in California (VAFB SLC-4E).

Launch Site Names Begin with 'CCA'

Query SELECT Date, Booster_Version, Launch_Site, Payload, PAYLOAD_MASS__KG_, Customer, Landing_Outcome FROM SPACEXTABLE WHERE Launch_Site LIKE 'CCA%' LIMIT 5

Date	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS__KG_	Customer	Landing_Outcome
2010-06-04	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	SpaceX	Failure (parachute)
2010-12-08	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	NASA (COTS) NRO	Failure (parachute)
2012-05-22	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	NASA (COTS)	No attempt
2012-10-08	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	NASA (CRS)	No attempt
2013-03-01	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	NASA (CRS)	No attempt

EXPLANATION

LIKE 'CCA%' matches sites starting with CCA and LIMIT 5 keeps the first five rows: the earliest Falcon 9 flights, all from CCAFS LC-40, carrying Dragon test and CRS missions with no successful landing.

Total Payload Mass

Query `SELECT SUM(PAYLOAD_MASS__KG_) AS Total_Payload_Mass FROM SPACEXTABLE WHERE Customer = 'NASA (CRS)'`

Total_Payload_Mass
45596

EXPLANATION

SUM over launches whose customer is 'NASA (CRS)': boosters carried 45,596 kg in total for NASA's Commercial Resupply Services missions.

Average Payload Mass by F9 v1.1

Query `SELECT AVG(PAYLOAD_MASS__KG_) AS Avg_Payload_Mass FROM SPACE_TABLE WHERE Booster_Version = 'F9 v1.1'`

Avg_Payload_Mass
2928.4

EXPLANATION

AVG over launches with booster version 'F9 v1.1': the average payload was 2,928.4 kg.

First Successful Ground Landing Date

Query `SELECT MIN(Date) AS First_Ground_Pad_Success FROM SPACEXTABLE WHERE Landing_Outcome = 'Success (ground pad)'`

First_Ground_Pad_Success
2015-12-22

EXPLANATION

MIN(Date) where Landing_Outcome = 'Success (ground pad)': the first successful ground-pad landing was on 22 December 2015.

Successful Drone Ship Landing with Payload between 4000 and 6000

Query `SELECT Booster_Version FROM SPACEXTABLE WHERE Landing_Outcome = 'Success (drone ship)' AND PAYLOAD_MASS__KG_ > 4000 AND PAYLOAD_MASS__KG_ < 6000`

Booster_Version
F9 FT B1022
F9 FT B1026
F9 FT B1021.2
F9 FT B1031.2

EXPLANATION

Four boosters landed on a drone ship while carrying 4,000–6,000 kg, all of the Full Thrust (FT) generation: B1022, B1026, B1021.2 and B1031.2.

Total Number of Successful and Failure Mission Outcomes

Query `SELECT Mission_Outcome, COUNT(*) AS Total FROM SPACEXTABLE GROUP BY Mission_Outcome`

Mission_Outcome	Total
Failure (in flight)	1
Success	98
Success	1
Success (payload status unclear)	1

EXPLANATION

GROUP BY Mission_Outcome with COUNT(*): 100 missions succeeded (98 + 1 with a trailing space + 1 with unclear payload status) and only 1 failed in flight.

Boosters Carried Maximum Payload

Query `SELECT DISTINCT Booster_Version FROM SPACEXTABLE WHERE PAYLOAD_MASS__KG_ = (SELECT MAX(PAYLOAD_MASS__KG_) FROM SPACEXTABLE)`

Booster_Version
F9 B5 B1048.4
F9 B5 B1049.4
F9 B5 B1051.3
F9 B5 B1056.4
F9 B5 B1048.5
F9 B5 B1051.4
F9 B5 B1049.5
F9 B5 B1060.2
F9 B5 B1058.3
F9 B5 B1051.6
F9 B5 B1060.3
F9 B5 B1049.7

EXPLANATION

A subquery finds the maximum payload (15,600 kg); 12 boosters carried it, all Block 5 (F9 B5) boosters.

2015 Launch Records

Query `SELECT substr(Date,6,2) AS Month, Landing_Outcome, Booster_Version, Launch_Site FROM SPACEXTABLE WHERE substr(Date,1,4)='2015' AND Landing_Outcome='Failure (drone ship)'`

Month	Landing_Outcome	Booster_Version	Launch_Site
01	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
04	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40

EXPLANATION

`substr(Date, 6, 2)` gives the month and `substr(Date, 1, 4) = '2015'` the year. Two drone-ship landings failed in 2015, in January (B1012) and April (B1015), both from CCAFS LC-40.

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

Query `SELECT Landing_Outcome, COUNT(*) AS Count FROM SPACEXTABLE WHERE Date BETWEEN '2010-06-04' AND '2017-03-20' GROUP BY Landing_Outcome ORDER BY Count DESC`

Landing_Outcome	Count
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

EXPLANATION

GROUP BY landing outcome, ORDER BY count DESC. "No attempt" leads (10); drone-ship successes and failures tie at 5 each; ground-pad successes (3) were still rare in this early period.

03

Launch Sites Proximity Analysis

Interactive maps built with Folium

Folium Map — All Launch Sites

folium.Circle + folium.Marker at each site's latitude / longitude

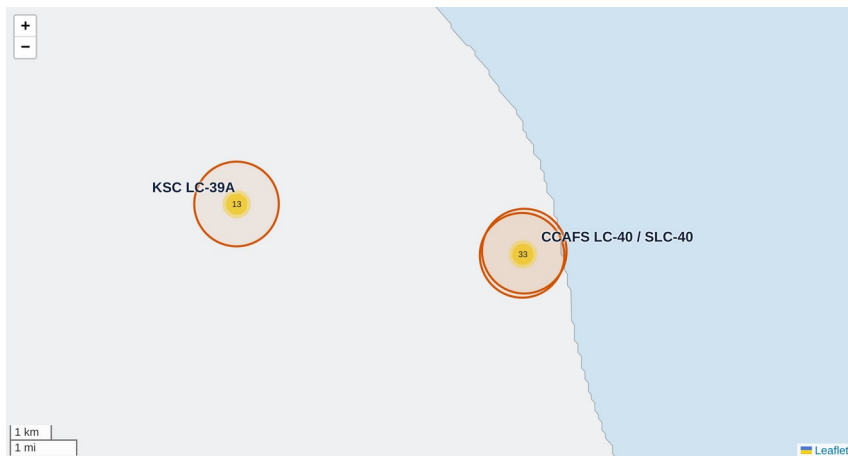


FINDINGS

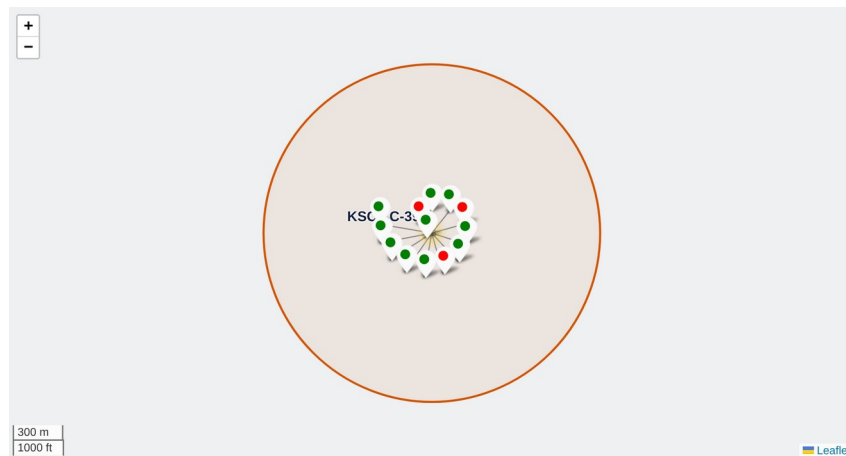
All four sites sit on a coast at the southern edge of the U.S. (close to the Equator). Florida sites launch east over the Atlantic; Vandenberg launches south over the Pacific for polar orbits.

Folium Map — Launch Outcomes per Site

MarkerCluster of every launch: green marker = landed (class 1), red = failed (class 0)



Clusters show 13 launches at KSC LC-39A and 33 at the CCAFS pad



Clicking the KSC LC-39A cluster: 10 green, 3 red markers

FINDINGS KSC LC-39A has the best ratio (10 landed / 3 failed, 76.9%). The CCAFS pad (LC-40 + SLC-40) flew most launches but mostly early ones, so it has the most failures (10 / 23).

Folium Map — Proximity of CCAFS SLC-40

folium.PolyLine from the launch site, distances computed with the haversine formula



Nearest...	Distance
Coastline	0.87 km
Titusville	23.2 km
Cocoa	25.5 km
Melbourne	53.4 km

FINDINGS

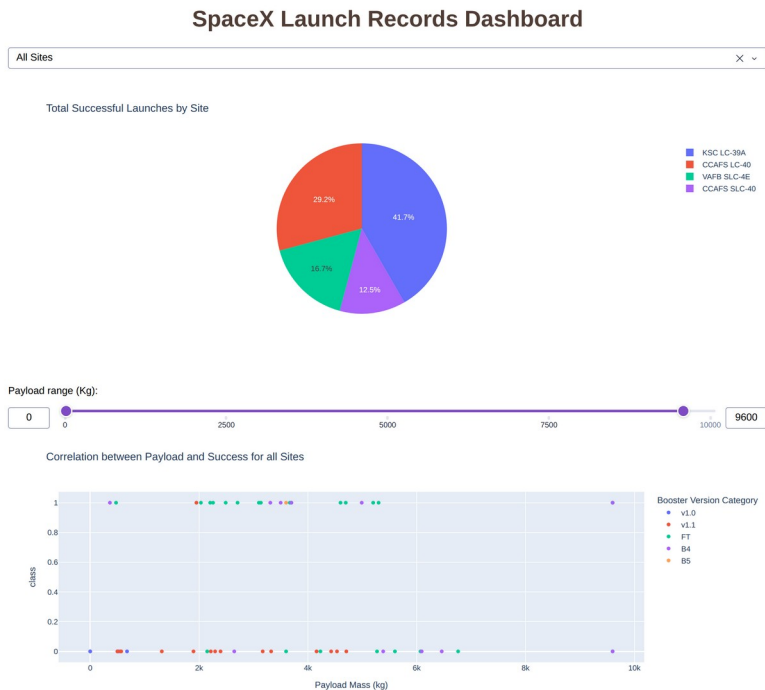
The pad is under 1 km from the sea (safe abort and recovery over water) and 23+ km from the nearest towns for public safety.

04

Build a Dashboard with Plotly Dash

Interactive launch-records dashboard (spacex_dash_app.py)

Dashboard Overview



Dropdown (site-dropdown): All Sites or one of the four launch sites.

Pie chart (success-pie-chart): all sites → share of successes per site; one site → success vs. failure.

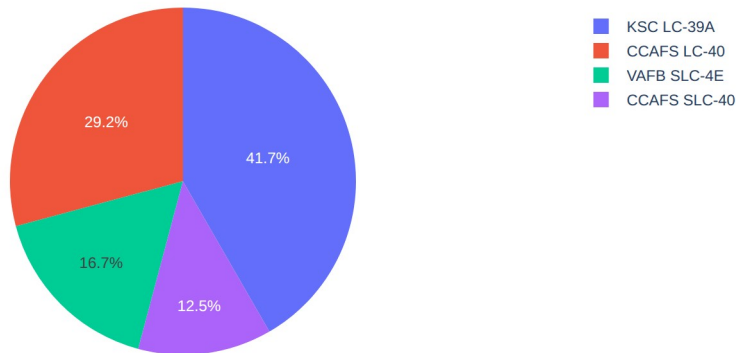
Range slider (payload-slider): filters payload from 0 to 10,000 kg in 1,000 kg steps.

Scatter (success-payload-scatter-chart): payload vs. class, coloured by booster version; reacts to both the dropdown and the slider (two Inputs in one callback).

Dashboard — Successful Launches by Site

Dropdown = All Sites

Total Successful Launches by Site



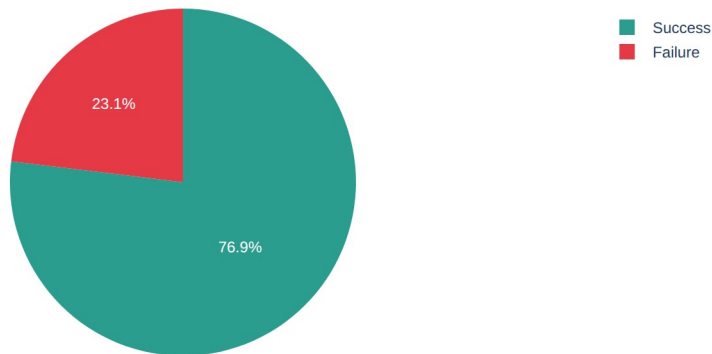
FINDINGS

KSC LC-39A has the largest share of successful launches (41.7%), then CCAFS LC-40 (29.2%), VAFB SLC-4E (16.7%) and CCAFS SLC-40 (12.5%). Shares partly reflect how many missions each site flew.

Dashboard — Site with the Highest Success Rate

Dropdown = KSC LC-39A

Success vs. Failed Launches for site KSC LC-39A



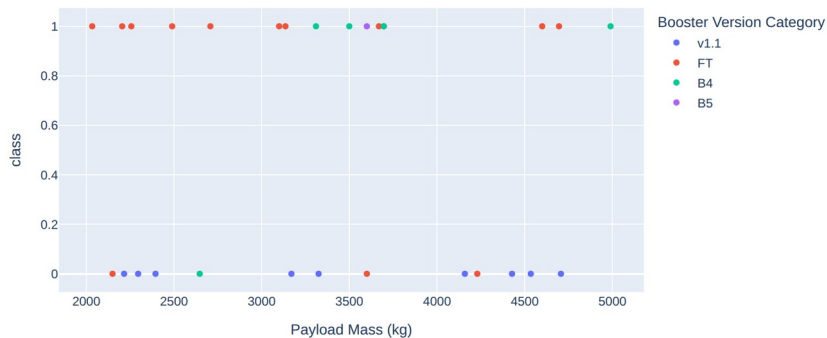
FINDINGS

KSC LC-39A landed 10 of 13 first stages: a 76.9% success rate, the best of any site (CCAFS LC-40 27%, CCAFS SLC-40 43%, VAFB SLC-4E 40%).

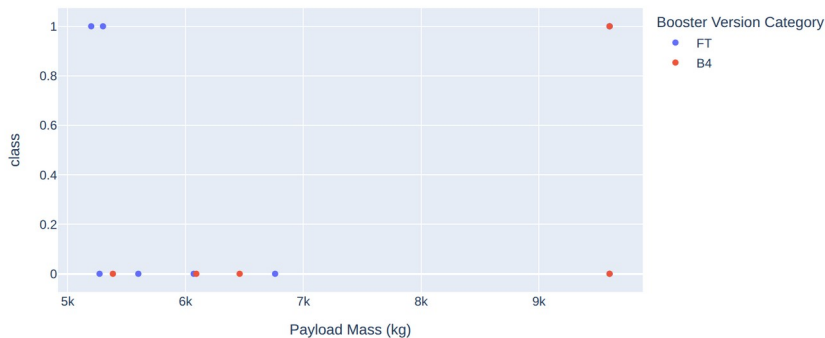
Dashboard — Payload vs. Launch Outcome

Same scatter, two slider settings (all sites)

Correlation between Payload and Success for all Sites



Correlation between Payload and Success for all Sites



Slider 2,000–5,000 kg: 16 of 29 landed (55%)

Slider 5,000–10,000 kg: 5 of 14 landed (36%)

Payload sweet spot: 2,000–5,000 kg has the most successes and the highest rate; below 2,000 kg only 23% landed (mostly early v1.0/v1.1 flights).

Booster matters most: FT boosters landed 16 of 24 (67%), B4 6 of 11; v1.1 landed only 1 of 15 and v1.0 none.

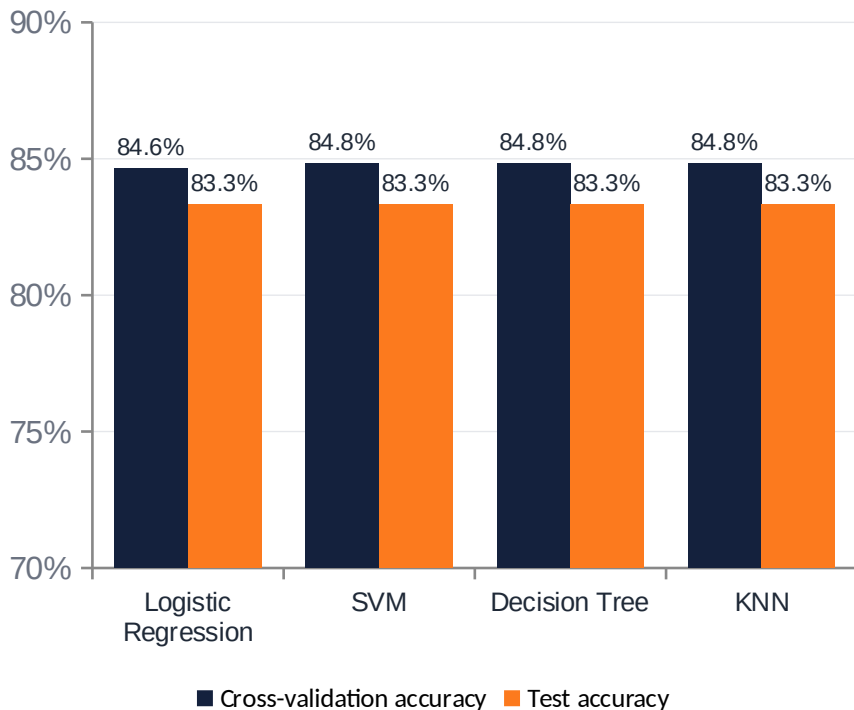
05

Predictive Analysis (Classification)

Classification results

Classification Accuracy

Bar chart of accuracy for all models. GridSearchCV (cv=10) on 72 training launches, scored on 18 test launches



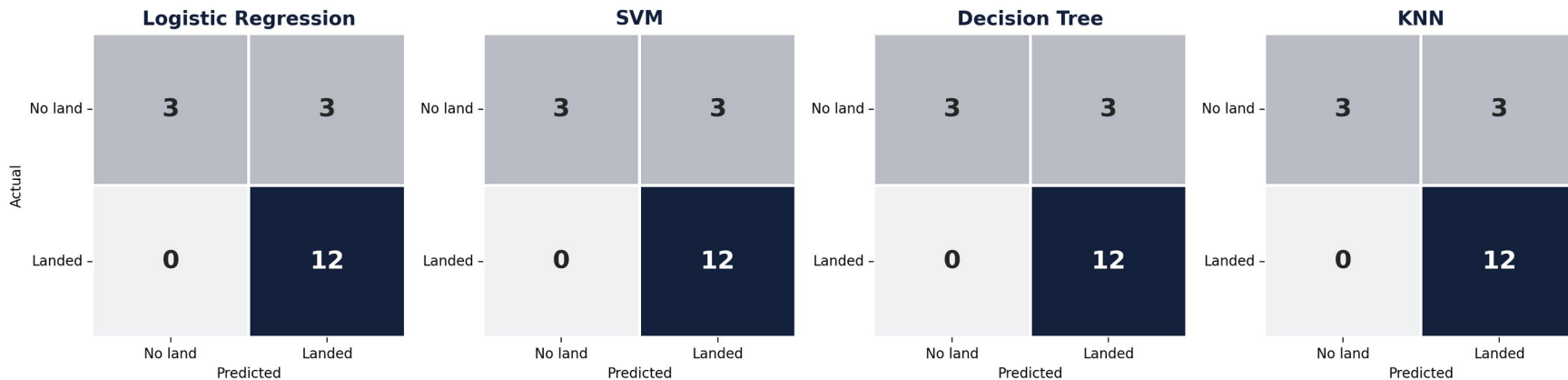
Model	CV	Test	Prec.	Recall	F1
Logistic Regression	84.6%	83.3%	80%	100%	88.9%
SVM	84.8%	83.3%	80%	100%	88.9%
Decision Tree	84.8%	83.3%	80%	100%	88.9%
KNN	84.8%	83.3%	80%	100%	88.9%

FINDINGS

On this split every model gets 15 of 18 right (83.3%), with 100% recall and 80% precision. SVM, decision tree and KNN share the top CV score (84.8%), so one split cannot separate them; see the robustness check on the next slides. Y-axis starts at 70%.

Confusion Matrix

Test set (18 launches) for all four tuned models

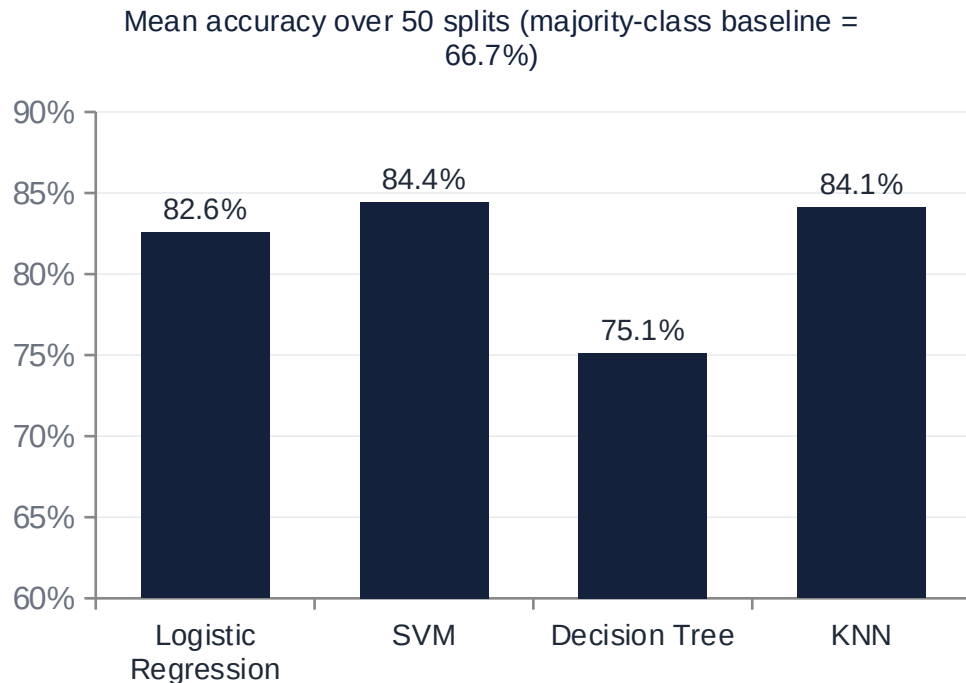


FINDINGS

All four models produce the same matrix: 12 true positives, 3 true negatives, 3 false positives and 0 false negatives. Every landing was caught, but 3 of the 6 failed landings were predicted to land, so the models are over-optimistic about failures.

Which Model Is Best? — Robustness Check

50 different train/test splits (5-fold × 10 repeats) using each model's tuned hyperparameters



BEST MODEL: SVM

SVM (sigmoid kernel) is the most accurate and stable: $84.4\% \pm 8.0\%$, ahead of KNN (84.1%) and logistic regression (82.6%). The decision tree drops to 75.1%: its single-split tie was luck, and it overfits. All models clearly beat the 66.7% baseline.

Conclusions

1 Experience pays off

Landing success rose from 0% before 2014 to 90% in 2019; flight number is the strongest single signal.

2 Site and orbit matter

KSC LC-39A has the best record (76.9%). GTO is the hardest orbit (52%); SSO, GEO, HEO and ES-L1 landed every time.

3 Hardware and payload

v1.0/v1.1 boosters rarely landed; FT, B4 and B5 account for nearly all successes. 2,000–5,000 kg payloads land most often.

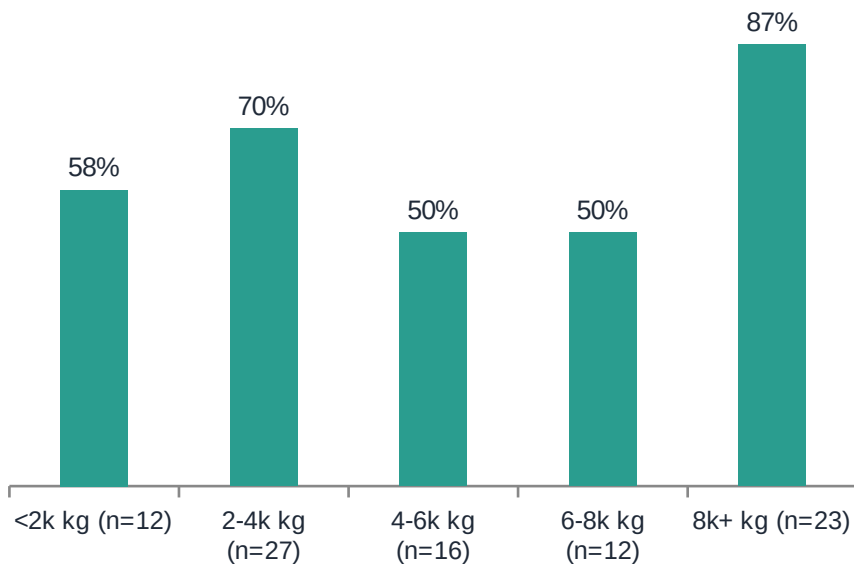
4 SVM is the best model

All models reach 83.3% on the test set; across 50 splits SVM is best (84.4%). Errors are false positives, so treat predicted landings with care.

Innovative Insights — Beyond the Template

What a competing bidder like Space Y should take away

Landing success by payload band



Payload is not a penalty: the heaviest band (8,000 kg+) lands most often (87%), because heavy missions flew late on mature boosters. Era confounds payload.

Price a GTO bid higher: GTO missions land only about half the time, so SpaceX's true cost on those launches is closer to expendable pricing.

One split is not enough: four models tied at 83.3% on the 18-launch test set, but resampling 50 splits showed SVM best and the decision tree overfitting (75%).

Next step: add launch year and booster block as explicit features and retrain on post-2020 launches to test whether accuracy holds.

Appendix

GitHub repository: <https://github.com/kautzarichramsya/Applied-Data-Science-Capstone>

Data sources: SpaceX REST API (IBM static snapshot `API_call_spacex_api.json`), Wikipedia "List of Falcon 9 and Falcon Heavy launches", IBM `Spacex.csv` and `spacex_launch_dash.csv`.

Best hyperparameters:

- Logistic regression: `C=0.01`, `penalty=l2`, `solver=lbgfs`
- SVM (best model): `kernel=sigmoid`, `C=1`, `gamma=0.0316`
- Decision tree: `criterion=gini`, `max_depth=8`, `min_samples_leaf=4`, `splitter=random`
- KNN: `n_neighbors=10`, `p=1`, `algorithm=auto`

Repository contents: notebooks for data collection (API), web scraping, data wrangling, EDA with SQL, EDA with visualization, Folium maps and machine learning prediction, plus `spacex_dash_app.py`.